

Activity Guide — middle / module-02-panchabhuta

IKS Curriculum

Table of Contents

Activity 01 — Element Sort

Module: Panchabhūta · **Band:** Middle · **Day:** 4 · **Time:** 25 min (within Day 4's 45 min)

Type

Hands-on classification, paired

Goal

Each pair sorts 5 real objects into the five-element framework and documents which other elements are also present in each.

Materials per pair

- 3 objects from each student's home (5–6 objects per pair after duplicates)
- 2 mystery objects supplied by the teacher per pair (e.g. a stretched balloon, a glass of cloudy water, a battery, a piece of camphor, a chunk of beeswax, a small mirror)
- 1 printed worksheet (next page)
- 1 pen each

Set-up (teacher, before class)

- Lay out 5 tray-areas around the room, each clearly labelled with one element name (Sanskrit + English).
- Distribute 2 mystery objects to each pair as they enter.

Instructions (student-facing — printable handout)

Element Sort — work in pairs.

1. Lay out your 5 objects on your desk.
2. For each object, decide which element is **dominant** (the strongest match). Place the object on the matching element tray.
3. Fill in the worksheet table below for all 5 objects.
4. **Disagree with your partner?** Argue it out before you fill in the row. If you can't agree, write both answers and circle the one your pair finally votes for.

5. **Gallery walk (last 5 min):** visit two other pairs' trays. If you disagree with where they placed an object, write a sticky note explaining why (kindly).

Worksheet (one per pair)

#	Object	Dominant element	Other elements present	Strongest <i>tanmātra</i>	Disagreement notes
1					
2					
3					
4					
5					

Discussion prompts (last 5 min, whole class)

- Which object was the hardest to classify? Why?
- Was there an object that two pairs put on different trays? What was the strongest argument for each side?
- Did any object seem to fit on *no* tray? What does that tell us?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners are pointing at the object, talking it through, sometimes disagreeing then resolving	“What about the smell?” / “Yes but the form is more obvious” / “Let’s say dominant is <i>prthvī</i> but <i>gandha</i> is the strongest <i>tanmātra</i> ”
Needs intervention	One partner doing all the writing while the other sits silently; rows filled in with one-word answers and no thought	Silence; “I don’t know, write whatever”

Safety / sensitivity

- No fire / no flame in this activity (despite *tejas* — students can find heat sources without lighting anything).
- No food sharing for tasting; if a student wants to demonstrate *rasa*, they taste their own food, not a classmate’s.
- Allergy check before allowing mystery objects like peanut shells, citrus peel, etc.

Differentiation

- **Tier up:** After sorting, the pair picks their single “most mixed” object and writes a paragraph allocating percentages to each element (e.g. 50% *pr̥thvī*, 30% *ap*, 15% *vāyu*, 5% *tejas*, 0% *ākāśa*). Defend the percentages.
- **Tier down:** Sort first into THREE categories (*pr̥thvī*, *ap*, *vāyu*) and come back to add *tejas* and *ākāśa* once the first three are confident.

Clean-up

- Each pair takes their own objects home.
- Teacher resets the 5 trays for the next class.
- Worksheet collected (stored in the project folder for Day 8 reference).

Activity 02 — Sense Station Rotation

Module: Panchabhūta · **Band:** Middle · **Day:** 3 · **Time:** 15 min (within Day 3’s 45 min)

Type

Hands-on sensory perception, paired

Goal

Students experience each *tanmātra* directly and notice when more than one perception is happening simultaneously.

Materials per group

5 stations (one per element), each set up with:

- **Station *ākāśa* (sound)** — a small bell or two metal spoons. A printed label: “*Perceive sound. Use your ears.*”
- **Station *vāyu* (touch)** — a piece of cotton wool, a small ice pack (or cold water in a sealed bag), a piece of sandpaper. Label: “*Perceive touch. Use your skin.*”
- **Station *tejas* (form / colour)** — a small mirror, a coloured plastic sheet, an unlit torch turned on. Label: “*Perceive form. Use your eyes.*”
- **Station *ap* (taste)** — a tiny pinch of salt, a tiny pinch of sugar (or jaggery), on individual paper plates. (Or a jar of mint leaves to chew if no food allowed.) Label: “*Perceive taste. Use your tongue.*”
- **Station *pr̥thvī* (smell)** — a sprig of mint, a slice of orange peel, a small jar of soil after wetting. Label: “*Perceive smell. Use your nose.*”

Set-up (teacher, before class)

- Arrange 5 stations around the room.
- Each pair is assigned a starting station (rotate clockwise every 2 min).
- 5 stations × 2 min each = 10 min for rotation + 5 min discussion.

Instructions (student-facing — printable handout)

Sense Station Rotation — work in pairs.

At each station, spend 2 minutes:

1. Use the sense organ named on the label to perceive the sample.
2. On your worksheet, write one sentence describing **what you perceived**.
3. Then ask yourself: **what OTHER sense could perceive something here at the same time?** Write that too.

Rotate when the teacher rings the bell.

Worksheet (one per pair)

Station	Tanmātra	What I perceived	Other element / tanmātra I noticed
ākāśa	<i>śabda</i> (sound)		
vāyu	<i>sparsā</i> (touch)		
tejas	<i>rūpa</i> (form)		
ap	<i>rasa</i> (taste)		
pr̥thvī	<i>gandha</i> (smell)		

Discussion prompts (last 3 min)

- At which station did you notice *more than one* element at once?
- Example: at the bell — did you HEAR the sound (*ākāśa*) and also FEEL the vibration in your fingers if you touched the bell (*vāyu*)?
- What does this tell us about how the elements actually appear in the world?

Safety / sensitivity

- **Diabetic check** before using sugar / jaggery — substitute with salt or a non-edible smell-only sample.
- **Allergy check** for peel, mint, soil.
- No sharing of food samples between students.
- Cold pack: don't let students hold it for more than 30 seconds.

Differentiation

- **Tier up:** At each station, name ONE everyday object that demonstrates the same tanmātra better. (E.g. ākāśa station — “I hear better with a singing bowl than this bell.”)
- **Tier down:** Just write what you perceived. Skip the “other element” column.

Clean-up

- Teacher collects all materials; students do not eat from leftover samples.

Activity 03 — Element Walk (Field Observation)

Module: Panchabhūta · **Band:** Middle · **Day:** 7 (debriefed in class; observation done at home overnight)

Type

Independent field observation, 20 min, completed at home or on the school grounds

Goal

Each student finds ONE example of an element imbalance in their neighbourhood and documents it.

Materials per student

- Notebook + pen, or a smartphone for photos
- The element-walk worksheet (below)

Instructions (student-facing — printable handout)

Element Walk

Spend 20 minutes walking around your neighbourhood (or the school grounds). Look for ONE place where one of the five elements is clearly out of balance — too much, or too little.

Examples of what to look for:

Element	Excess	Depletion
<i>pr̥thvī</i> (earth)	Pile of plastic waste; bald soil from no plants	Eroded soil; bare slope
<i>ap</i> (water)	Stagnant water pool (mosquitoes); flooded drain	Dry tap; cracked dry earth
<i>tejas</i> (fire/heat)	Heat-island parking lot at noon; smoke from burning garbage	Cold dark corner where nothing grows
<i>vāyu</i> (air)	Vehicle exhaust at a signal; dust storm	Stagnant indoor air; no breeze in a closed alley
<i>ākāśa</i> (space/sound)	Constant honking; loudspeakers	An eerily silent abandoned spot

Document on the worksheet:

Worksheet (one per student)

Date: _____ Time of observation: _____

Location (one line): _____

Which element is out of balance? _____

Is it in EXCESS or DEPLETION? (circle one) EXCESS / DEPLETION

What did you observe? (3 sentences)

What is one observable consequence?

What is ONE small action you (or your community) could take?

(Optional) Sketch or describe what you saw:

Discussion prompts (Day 7 in class)

- Whose observation was the most specific (named a real street, a real time)?
- Did any two students choose the same location? How did their observations differ?
- Which “small actions” feel most doable? Which feel hardest?

Safety / sensitivity

- **No solo walks for younger students (Grade 6).** Pair with a sibling, parent, or do the walk on school grounds during break.
- **Avoid actually inhaling smoke, drinking unsafe water, etc.** Observation is the goal — not exposure.
- **Be aware** of sensitivity around naming specific neighbourhoods as polluted (some students live there). Frame the observation as “everyone’s neighbourhood has imbalances, including mine.”

Differentiation

- **Tier up:** Find TWO related imbalances (e.g. polluted water leads to depleted plant life). Document the connection.

- **Tier down:** Just describe what you saw. The “one action” can be one of the suggestions from the in-class examples.

Clean-up

- Bring the worksheet to Day 7 class.
- Photos optional — only if devices are allowed in school per school policy.

Project Brief

Project Brief — Module 2 Panchabhūta (Middle)

Work sessions: Day 8 + Day 9 (in class) · **Presentation + summative:** Day 10 · **Total marks:** 40

The project

Each pair (or individual, with teacher approval) chooses **one** of the five elements and produces a small project that shows:

1. **What the *pañcabhūta* tradition says** about this element (its tanmātra, its position in the creation sequence, the quality-count rule, what it “does”). Must cite at least one source — a verse from the RAG cache, or a reference from `sources.md`.
2. **One modern-science parallel** — a concept from chemistry, biology, physics, or ecology that overlaps with this element. Explain where the parallel works and where it breaks down.
3. **One local imbalance** — a real example in your neighbourhood where this element is in excess or depletion, with one doable action.

Why this matters

You’re learning to hold two intellectual frameworks side-by-side without forcing one to win. This is a core skill — not just for *pañcabhūta*, but for any time you encounter a way of seeing the world that’s different from your own.

What success looks like

A strong project will: - Demonstrate the module’s central concepts accurately (tanmātra, creation sequence, quality-count rule). - Connect the Indian tradition to a modern scientific concept fairly — without saying “ancient Indians knew this already.” - Show your own thinking, not just summary. - Be clearly communicated in your chosen medium.

Format options

Format	Best for elements	Notes
Poster (A2 or larger)	Any	Lots of visual space; works for

Format	Best for elements	Notes
		nervous presenters
Live demonstration (2 min hands-on)	<i>tejas</i> (fire), <i>vāyu</i> (air), <i>ap</i> (water)	Needs safety check
Short written story (1 page)	<i>ākāśa</i> , any	Read aloud or hand out
90-second video	Any	Pre-made and played in class
Infographic	<i>pṛthvī</i> (data-heavy), <i>ap</i>	Combine drawing + facts
Original song or poem	<i>vāyu</i> , <i>ākāśa</i>	90 sec performed live

Other formats with teacher approval. Whatever the format, the 3-minute time limit on Day 10 applies.

Working in pairs (or individually)

- **Pair work is encouraged**, especially for video/song/demo formats.
- **Individual is allowed**, especially if the topic is *ākāśa* (often suits a solo reflective piece).
- Groups of 3 are acceptable but each member must have a distinct sub-role (visuals / script / presenter, or research / draft / final).

Schedule

Day	What you do	Submit at end of class
5	Choose element + format + partner	Sticky note on the board
6	(Homework) Submit project plan paragraph	Project plan paragraph
8	Plan + start first artefact	Draft of first artefact (photo or scan accepted)
9	Refine + rehearse	Draft of final artefact
10	Present (3 min) + Q&A (1 min)	Final artefact + reflection slip

Citation rules

- At minimum **one** citation must appear in your project — visible on the poster, named in the video, or printed in the story.
- The citation must be from a source you can point to: a verse from one of the day-NN.json cache files in `.rag-cache/`, OR a reference from `sources.md`.
- If you invent a citation or pull one off the internet without verification, you'll score 0 on both **Use of sources** and **Originality**.

Rubric

See `rubric.md`. Read it before Day 8.

Self-reflection (submitted on Day 10 with project)

In 4–6 sentences, answer:

1. What did you set out to learn or make, and what did you actually end up with?
2. What is the most surprising thing you discovered while making this?
3. If you had three more days, what would you do differently?

Exemplars

A “Meets” exemplar will be added to `exports/pdf/middle-module-02-panchabhuta/exemplar-strong.pdf` after the first cohort presents.